Exact row multipliers beyond powers of two

Independent exploration for OMS — 15 October 2026

Decision. A complete exact-product selector and a finite export-aware selector are implemented and checked. They improve coefficient centering and avoid some abstentions in constructed narrow intervals. Neither admits any additional contract in the three requested physical scenarios. There are no new solver calls, performance claims or manuscript additions.

1. Complete exact-product selection

Write every nonzero row datum, including the right-hand side, as $z = \mathrm{sign}(z) N_z 2^{p_z}$, with N_z odd. Write a positive binary64 factor uniquely as $\alpha = h2^k$, h odd. Assuming gradual underflow, all products are exactly representable if and only if, for every datum,

$$hN_z \le 2^{53} - 1, \qquad p_z + k \ge -1074, \qquad p_z + k + \lfloor \log_2(hN_z) \rfloor \le 1023,$$

and the factor itself is representable. Thus $h \le \lfloor (2^{53} - 1)/\max_z N_z \rfloor$. Exponent and interval limits leave an interval of odd integers h at each exponent. For

$$Q(\alpha) = \max\{\alpha M, 1/(\alpha m)\}, \qquad m = \min_{a_j \neq 0} |a_j|, \quad M = \max_j |a_j|,$$

the endpoints and the neighbors of the real optimum suffice. Exact rational comparisons produce a global minimum on the full exact-export set. The power-of-two control restricts $h = 1$. These are elementary consequences of binary representation; historical novelty is not claimed. Q is the exponential of the logarithmic centering objective, not a condition number.

2. Reserving rounded-export error

For $\widehat{a} = \mathrm{fl}(\alpha a)$ and $\widehat{b} = \mathrm{fl}(\alpha b)$, compute exactly on the stored box X

$$\rho(\alpha) = \max_{x \in X} \left| (a - \widehat{a}/\alpha)^T x + \widehat{b}/\alpha - b \right|.$$

An exported residual at most ϵ and an integer-rounding allowance E give the sufficient original-row guarantee $\epsilon/\alpha + E + \rho(\alpha) \le \delta$. The exported residual is a hypothesis to verify, not a consequence of setting a solver tolerance. A nonzero export error on an unbounded coordinate makes this box certificate inconclusive; other model rows might supply bounds.

The finite *Aware* pool contains full exact GM, power-of-two GM, binary64 GM, inward endpoints, and eight adjacent floats on either side of binary64 GM. Candidates must meet the full contract and actual exported coefficient limits. This finite search is not globally optimal among rounded exports. Original rows, columns, bounds, objectives and integrality are fixed.

3. Validation scope

Three known SG–Ter–Mer models retain 1,440 integer variables each. This is one family, not external validation. The complete selector is checked by excluding every strictly better factor with a separate feasibility program. Five tests, 14,000 exhaustive toy-format comparisons and 300 binary64 boundary pairs supplement the original-data replay. No optimizer is imported or run.

4. Real contracts: no additional admissibility

All 30,240 rows are audited. Each physical scenario covers 12,960 rows, including the previously documented activation extension. The coefficient window is $[10^{-9}, 10^{9}]$, $\epsilon = 10^{-6}$, and $E = 10^{-9} \sum_{j \in J} |a_j|$, using the parent's binary64 inputs. Physical budgets are converted to each row's algebraic units. They are study accuracy targets, not operating standards.

Scenario (MW, m^3)	Rows	Pow2 passes	Exact passes	Aware passes
Demanding (.01, 100)	12960	12960	12960	12960
Reference (.1, 1000)	12960	12960	12960	12960
Permissive (1, 10000)	12960	12960	12960	12960

Without budgets, full exact GM changes 5,364/30,240 factors (17.74%) while retaining zero export defect. The median $Q_{\mathrm{Exact}}/Q_{\mathrm{Pow2}}$ over all rows is 1; the best ratio is 0.70710678. For example, a stored height row changes from factor 32 to 36 with all products exact. Each physical scenario changes 2,160/12,960 factors but rescues no contract.

In 17,773 rows, the only possible exact multipliers are powers of two, even before interval limits. In other rows, short odd significands permit more choices. Arbitrary re-emission can change this set: full invariance under arbitrary changes of units is not established for the discrete selector.

Binary64 GM without an export reserve also passes every physical contract. The maximum export defect divided by remaining budget is $1.732 \cdot 10^{-8}$ in the demanding scenario, $1.535 \cdot 10^{-9}$ in the reference scenario, and $1.517 \cdot 10^{-10}$ in the permissive scenario. Reserving this defect is therefore nonbinding here. Over the full unbudgeted bank, 4,320 cost-envelope rows have an unbounded box defect for rounded GM because affected coordinates lack finite stored bounds. This is a limitation of the box certificate, not proof of unbounded error over the feasible region. Exact exports avoid it.

5. Constructed narrow-interval sensitivity

For each physical row set $\delta = E + \epsilon/(\theta H)$, $H = 10^9/M$. These artificial targets are separate from application scenarios.

θ	Rows	Pow2 admits	Exact admits	Abstentions avoided
1/2	12960	12960	12960	0
3/4	12960	5400	8640	3240
7/8	12960	1080	4320	3240
15/16	12960	1080	4320	3240
1	12960	0	3240	3240

At $\theta = 1$, rows with coefficient magnitudes 1 require factor 10^9, which is binary64-exact but not a power of two. This rescues 3,240 power rows. Their residual target is around 10^{-15} MW, far below the requested 0.01 MW. This demonstrates a grid limitation, not a practical dispatch need or an incompatibility of the real-factor interval. The manuscript already distinguishes exact-grid abstention from real-interval incompatibility.

6. Joint limits: what the literature review establishes

The earlier joint-limits note treats a fixed nonsingular square matrix in the infinity norm, with fixed columns and factor boxes. It does not certify the solver's internally scaled matrix, all LP bases, or general MILP performance. Its unrestricted minimum and Skeel invariance are classical; see Higham (1994), equations 2.11–2.12. They must not be presented as discoveries.

The new literature check examined Charnes–Cooper (1962), pp. 181–184. Homogenizing positive fractional variables into linear constraints is an established technique. The previous substitution $v = t/\alpha$ is an instance of that approach, not a new general optimization method. This does not imply that the 1962 article solves the particular boxed row-scaling problem.

Qu et al., sections 1, 2.1 and 3.1 study optimal diagonal preconditioning in the spectral norm through quasiconvex and SDP reformulations. Their formulations include diagonal restrictions, but do not prove our boxed infinity-norm formula. The historical priority of that specific extension remains unestablished; a limited search cannot settle it. More importantly, no real instance has shown that the joint certificate changes a useful dispatch decision.

For the present exact-export implementation, Goldberg (1991) provides the standard binary-arithmetic background. Berthold–Hendel (2021), p. 3662, explicitly describe power-of-two scaling to avoid data error. These precedents support the controls and limit novelty claims. Neither paper is cited as a proof of our complete discrete selector.

7. Reproduction and editorial decision

The independent replay checks 246,240 factor decisions, 852,075 products, 138,240 discrete optimality conditions, and 64,800 constructed intervals.

Canonical model inputs and exact decisions are included in the package. Their identity to original LP exports was checked in the pinned parent study; this study does not re-import LPs. A derived counter was corrected after the first run: the odd-significand limit 2 also permits only 1. Factor decisions and stress outputs retained identical hashes. The original source, correction note and before/after hashes are retained.

The reported selection time includes caching, metadata and rational checks; it is not MILP solution work. Complete replay requires Python, pytest and pdfLaTeX, without optimizer libraries or licences. The frozen OMS v1.3.5 delivery and ten earlier research packages are checked unchanged by SHA256.

Conclusion for this exploration. Full exact multipliers give a verified software extension and fewer grid abstentions under constructed targets. The requested physical contracts show no additional admissibility. There is no evidence here for better MILP performance or for another OMS contribution. No solver campaign is launched and the manuscript is unchanged. The original-integer LP–DW diagnosis and a validated predictive selector remain separate outstanding tasks.